

LOIS KOSTNER

MSc student in Aerospace Engineering | CFD, Aero-Thermodynamics & Turbomachinery

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EDUCATION

MSc in Aerospace Engineering

Starting October 2026 - Expected 2028

- University: Technische Universität München

BSc in Aerospace Engineering

September 2022 – March 2026

- University: Alma Mater Studiorum – University of Bologna
- Final grade: 110/110
- Thesis topic: The aerodynamic impact of the propulsive jet on a sounding rocket, using CFD simulations to assess variations in the main force and moment coefficients

SELECTED PROJECTS

Aerothermal CFD Analysis of the NASA C3X Turbine Vane

September 2026 – Ongoing

- Reconstructed the C3X vane profile from the NASA CR-174827 data and prepared a periodic 2.5D computational domain consistent with the experimental configuration.
- Currently developing a fully structured multi-block mesh in ANSYS ICEM CFD, with near-wall refinement aimed at resolving the thermal boundary layer.
- Preparing a RANS CFD workflow in ANSYS Fluent for prediction of the external heat-transfer distribution and comparison with published NASA measurements.

CFD Analysis of the Propulsive Jet Effect on the Aerodynamic Coefficients of a Sounding Rocket

October 2025 – March 2026

- Quantified the impact of the exhaust jet on the aerodynamic force and moment coefficients of a sounding rocket.
- Defined and verified the numerical methodology through mesh-independence, domain-independence, and solver-sensitivity studies.
- Ran ANSYS Fluent simulations for both powered and non-powered configurations, supported by convergence checks and verification analyses.
- Compared the two configurations in terms of lift, drag, and pitching moment, and assessed the associated numerical uncertainty.

CFD-FEM Aero-Structural Analysis of a Sounding Rocket

July 2025 – October 2025

- Conducted the aero-structural assessment of a sounding rocket under the most critical flight condition, combining peak aerodynamic loads with maximum motor thrust.
- Developed and implemented the CFD-FEM coupling workflow adopted in the EuRoC Technical Report.
- Computed distributed aerodynamic loads in ANSYS Fluent, built an equivalent beam model, and reconstructed the internal load path through an inertia-relief approach.
- Applied the resulting loads to the main structural components and verified structural integrity through safety-margin assessment.

EXPERIENCE

Co-founder and Vice-President, Aurora Rocketry

March 2024 – October 2025

- Co-founded a student rocketry association at the University of Bologna and helped scale it to more than 100 members.
- Managed the association's financial operations, including cash flow, expense tracking, payments, and budget organization.
- Contributed to sponsor relations, institutional communication, and the preparation of technical and presentation material.
- Helped secure €20,000 in PNRR funding to support the team's activities.
- Contributed to the team's EuRoC 2025 application and competition preparation for the sounding rocket prototype *Nemesis*.

TECHNICAL SKILLS

SolidWorks, ANSYS SpaceClaim: geometry creation, simplification, cleanup, and preparation for CFD meshing; equivalent beam model development for structural analysis.

ANSYS Fluent Meshing, ANSYS ICEM CFD: unstructured poly-hexcore meshing with prism boundary layers; developing proficiency in structured multi-block meshing with ICEM CFD; mesh quality assessment and near-wall refinement.

ANSYS Fluent: simulation setup, verification, mesh/domain sensitivity studies, convergence assessment, aerodynamic load evaluation, and post-processing.

ANSYS Static Structural, ANSYS Modal: structural modelling of load-bearing components, equivalent beam modelling, and modal analysis of aerospace structures.

MATLAB: modular post-processing tools, automated data handling, and custom analysis scripts for simulation workflows.

LANGUAGES

Italian – Native

Ladin – Native

German – B2 Goethe-Zertifikat

English – 7.5 IELTS overall score